

# Giridhar G

## Embedded Hardware Engineer

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### PROFILE

Embedded Hardware Design Engineer with 3 years of experience in designing embedded and communication products including IoT gateways, data concentrators, Ethernet switches, media converters, and industrial interface modules. Strong expertise in digital electronics, schematic design, 2-layer and 4-layer PCB layout, Ethernet, serial, cellular interfaces, and power supply design. Experienced in board bring-up, production support, and cross-functional collaboration to deliver reliable, production-ready hardware.

### SKILLS

#### Digital & Embedded Hardware

Digital Electronics, Embedded Hardware Design, Schematic Capture, PCB Layout (2L/4L), Stack-up Planning, EMI/EMC Practices, Board Bring-up, Debugging

#### Power Electronics

AC-DC Flyback (TNY278PN), DC-DC (9–60 V → 5 V, TPS54560), Power Tree Design, Protection Circuits

#### Communication & Networking

RS-232, RS-485, Ethernet (10/100 & Gigabit), Fiber Optics (SM/MM), USB 2.0, I<sup>2</sup>C, SPI, CAN, PLC

#### Industrial I/O

DI/DO, Analog I/O, RTU Hardware Design, Isolation Techniques

#### Cellular & Wireless

4G LTE (Quectel EC25, EC200), Dual-SIM Hardware Design, OpenWRT Platforms, Wi-Fi, Bluetooth

#### Manufacturing & Tools

OrCAD, SolidWorks (3D PCB), Gerbers (RS-274X), BOM, DFM/DFT, Oscilloscope, Logic Analyzer

### PROFESSIONAL EXPERIENCE

#### Creative Micro Systems

Embedded Hardware Design Engineer

02/2023 – Present

Bengaluru

- Worked on end-to-end **embedded hardware development** for industrial, communication, and IoT systems, from concept to production.
- Responsible for **schematic design, netlist generation, and multi-layer PCB design (2-layer & 4-layer)** with stack-up planning and EMI-aware digital layout practices.
- Defined **system-level hardware architecture**, including processor/module selection, interface planning, power tree design, and protection circuits.
- Designed and integrated **digital communication interfaces** such as serial, Ethernet, cellular, and field interfaces for reliable data communication.
- Developed supporting **power supply subsystems**, including isolated AC-DC and wide-input DC-DC converters for industrial applications.
- Performed **component selection and validation** considering performance, availability, cost, and manufacturing constraints.
- Led **hardware bring-up, testing, and debugging**, collaborating closely with firmware teams to resolve power, clocking, reset, and interface issues.

- Generated **production-ready manufacturing deliverables** including Gerbers, BOMs, and assembly documentation, and supported prototype to production stages.
- Coordinated with mechanical teams using **3D PCB models** to ensure enclosure fit and system-level integration.

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## PROJECTS

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- Designed **Data Concentrator Units** for meter data collection using isolated **RS-232, RS-485, Ethernet, and PLC modules** with MYC-Y6ULG2 processor module.
- Developed an **Intelligent 4G Industrial Router** with **dual-SIM support**, integrating Atheros-based routing module and **Quectel EC25/EC200** LTE modems.
- Designed an **IoT Gateway** with **dual-SIM 4G, Wi-Fi, Bluetooth, RS-232/RS-485, and 4 DI / 2 DO** interfaces based on **OpenWRT** platform.
- Developed a **cellular meter interface modem** with **dual-SIM LTE** and **RS-232 interface** for transformer and meter data acquisition using **Quectel EC25**.
- Designed **interface converter boards** including **RS-232 to RS-485** and **TTL to RS-232** for industrial communication systems.
- Developed **Ethernet media converters** supporting **Fast Ethernet and Gigabit Ethernet**, including **copper-to-fiber** and **multimode-to-single-mode fiber** conversion.
- Designed **isolated AC-DC flyback power supplies** using **TNY278PN** and **wide-input DC-DC converters (9–60 V → 5 V)** using **TPS54560**.
- Developed **Digital and Analog I/O cards** for **RTU systems** using **STM32 microcontrollers**, supporting industrial field signals.

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## EDUCATION

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|-------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Coimbatore Institute of Technology</b>  | 07/2019 – 04/2023 |
| <i>BE in Electrical and Electronics Engineering</i>                                                                           | Coimbatore, India |
| <b>Velammal Matric Hr Sec School</b>       | 2019              |
| <i>HSC</i>                                                                                                                    | Theni, India      |

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## LANGUAGES

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|         |        |
|---------|--------|
| English | Tamil  |
| Kannada | German |